Lucky Verma

Machine Learning Engineer | Agentic Systems | Knowledge Graphs | RAG

667–910–4149 | luckie.verma30@gmail.com | LinkedIn | luckyverma.com | GitHub | Nashville Metro, TN

Summary

Machine Learning Engineer with 4+ years building production agentic AI systems, knowledge graphs, and deep learning pipelines for large-scale data infrastructure. Expert in multi-agent orchestration (LangGraph, DSPy), RAG pipelines, entity resolution, model evaluation, and real-time inference serving on GPU clusters.

Technical Skills

AI / LLM	**RAG**, Agentic AI, **LangGraph**, **DSPy**, LangChain, vLLM, Fine-tuning (LoRA, RLHF/DPO), Tool Use, HF Transformers
ML / DL	**PyTorch**, TensorFlow, FSDP, DeepSpeed, SLURM Multi-GPU, Triton Kernels, **Deep Learning**, **Computer Vision**, NLP, **CUDA**
Infra	**Knowledge Graphs** (Apache AGE, LightRAG), PostgreSQL/pgvector, **FastAPI**, **Docker**, K8s, Terraform, **AWS**, Redis, Model Monitoring
Languages	**Python**, SQL, C++, JavaScript, Bash

Work Experience

Eccalon LLC (AI and federal data systems) — *Nashville, TN*
Machine Learning Engineer — *Jun 2022 – Present*

- Architected and deployed production **deep learning** + agentic AI workflows for regulated document analysis, evidence retrieval, and corporate ownership reasoning using **LangGraph** orchestration, **DSPy** optimization, and multi-step reasoning across federal and corporate data sources.
- Engineered **knowledge graph** infrastructure (PostgreSQL + Apache AGE) with entity resolution across SAM.gov, SEC EDGAR, and GLEIF corporate registries, supporting multi-hop ownership-chain tracing with low-latency query response.
- Built production **document intelligence pipeline** with GPU-accelerated embeddings (**Ollama** + NIM fallback), LightRAG/PostgreSQL-backed retrieval, and source-grounded synthesis across federal databases and regulatory documents.
- Built **model evaluation** framework for LLM baselines, feature engineering, contamination checks, and multi-annotator validation in ownership and compliance-oriented reasoning workflows.
- Deployed production **RAG pipelines** on AWS with multi-path reasoning, hybrid vector/keyword search, and source-grounded retrieval; built GPU-accelerated CV pipelines with **PyTorch** and CUDA.
- Owned end-to-end **ML infrastructure** — Docker multi-service orchestration, Terraform IaC, **CI/CD**, **model serving**, **monitoring**, and multi-backend embedding fallback for production ML services.

University of Maryland, Baltimore County — *Baltimore, MD*
Graduate Research Assistant — *Jan 2022 – May 2023*

- Developed ensemble **1D U-Net** for biomedical signal classification (PPG → sleep staging), achieving **92%** accuracy; built web-based data science learning platform using CV/ML pipelines with **pandas** and **Streamlit**.

Vast Dream Group — *Mumbai, India (Remote)*
AI Specialist — *Jan 2021 – Aug 2021*

- Built recommendation engine using **BART** zero-shot classification (96% accuracy on MNLI); architected AI measurement systems for optics industry, contributing to **$1M+** revenue pipeline across 3 product lines.

TandM Techlabs — *Secunderabad, India*
Software Developer — *Jun 2018 – Dec 2020*

- Built urban mobility platform with geospatial ML: **Pandana** accessibility metrics, **Valhalla/OSMnx** route optimization on **PostGIS**, cached with **Redis** and containerized on **Kubernetes** — serving real-time discovery for B2B logistics.

Key Projects & Open Source

Ownership Risk Evaluation Workbench
Multi-source entity resolution | LLM baseline evaluation | Evidence tracing

- Built an evaluation workbench for corporate ownership reasoning, evidence tracing, and baseline comparison under regulated review workflows. Public descriptions are scoped to non-sensitive architecture and released validation artifacts.

InfiniteContext-1B: Million-Token LLM Architecture — *GitHub*
DeepSeek-V3 MLA | Distributed FSDP | Consumer GPU inference

- Building an experimental MLA-based long-context LLM architecture workbench with custom Triton kernels, FSDP distributed training plumbing, and reproducible evaluation scripts for consumer-GPU inference studies.

Education

University of Maryland, Baltimore County — *Baltimore, MD*
Master of Science in Computer Science – GPA: 3.97 — *May 2023*

- **DARPA Catalyst Award** ($50K) – Caselet, selected from **250+ teams** for AI-powered national security workforce tools

SRM University — *Chennai, India*
Bachelor of Technology in Electrical & Electronics Engineering – GPA: 3.9 — *May 2019*